JOHNS HOPKINS
WHITING SCHOOL of ENGINEERING

Executive and Professional Education

Great Learning



CERTIFICATE PROGRAM IN AGENTIC AI

14 WEEKS ONLINE | LIVE MASTERCLASSES BY JHU FACULTY | LIVE MENTORSHIP SESSIONS BY INDUSTRY EXPERTS

Tools and Techniques Covered:

python	OpenAI	Codex	LangChain	LangGraph
ReAct Reasoning and Acting	MCP	A2A	DeepEval.	ragas
DSPy	Claude	CHROMADB	tavily	

JOHNS HOPKINS UNIVERSITY ADVANTAGE

Johns Hopkins University (JHU) is a prestigious institution consistently ranked among the top research universities in the United States, with a strong reputation in research and education. JHU is well-positioned to offer compelling differentiators for its professional and continuing education programs.

WORLD-CLASS FACULTY EXPERTISE

The programs are taught by faculty who are distinguished in academia and have industry experience leading Artificial Intelligence practices at Fortune 500 companies. This blend of academic rigor and industry expertise offers unique insights into Agentic AI.

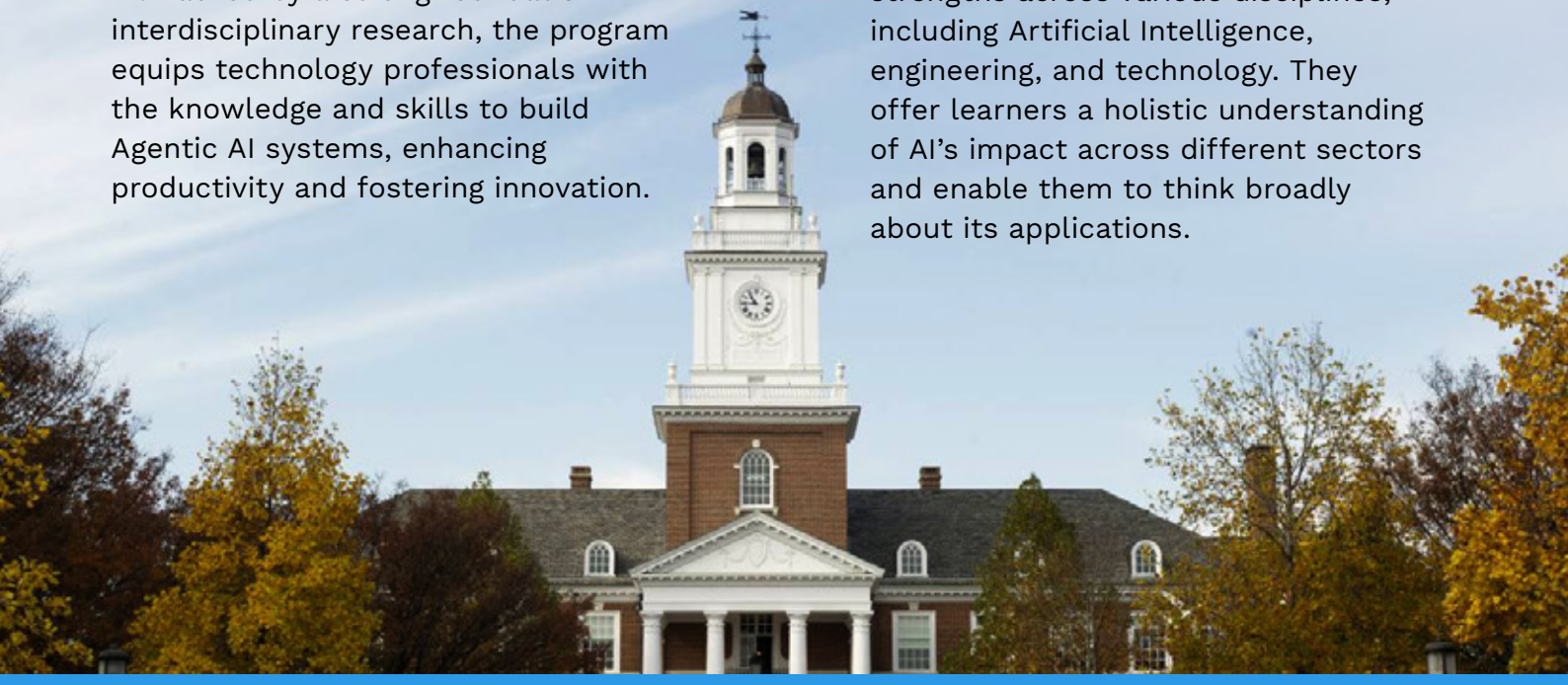
Johns Hopkins University stands at the forefront of AI innovation, with faculty who are recognized leaders in Artificial Intelligence, Agentic AI, Machine Learning, Deep Learning, and Generative AI. Backed by a strong foundation in interdisciplinary research, the program equips technology professionals with the knowledge and skills to build Agentic AI systems, enhancing productivity and fostering innovation.

RESEARCH-DRIVEN CURRICULUM

The curriculum is designed around the latest research and developments in Agentic AI, ensuring learners engage with current, relevant, and rigorously developed content. Johns Hopkins University is a leading institution in research and has maintained a top position in R&D for decades.

INTERDISCIPLINARY APPROACH

The programs leverage JHU's strengths across various disciplines, including Artificial Intelligence, engineering, and technology. They offer learners a holistic understanding of AI's impact across different sectors and enable them to think broadly about its applications.



#9
NATIONAL UNIVERSITY
U.S. NEWS & WORLD REPORT 2027

#2
COMPUTER INFORMATION TECHNOLOGY
U.S. NEWS & WORLD REPORT 2027

#17
BEST GLOBAL UNIVERSITY
U.S. NEWS & WORLD REPORT 2027

71% OF ORGANIZATIONS USE AI AGENTS. ONLY 11% OF AGENTIC PILOTS HAVE REACHED PRODUCTION.

SKILL GAP

88%

of organizations running AI agents have experienced a security incident, commonly caused by poor agent design and excessive permissions.

Gravitee, State of AI Agent Security, 2026

JUDGMENT GAP

Over

40%

of Agentic AI projects will be canceled by the end of 2027 due to escalating costs, unclear business value, or inadequate risk controls.

Gartner, 2025

GOVERNANCE GAP

Only

21%

of organizations have a mature governance model for Agentic AI.

Deloitte, State of AI in the Enterprise 2026

DESIGN, BUILD, AND EVALUATE AGENTIC AI SYSTEMS

The 14-Week **Certificate Program in Agentic AI** by Johns Hopkins University is designed for **data scientists, AI/ML engineers, and technology professionals** seeking to develop agentic systems that perform beyond the prototype stage. It is equally relevant to **technical and product managers** championing Agentic AI adoption within their teams, providing the technical grounding required to guide design decisions.

This program offers a structured progression from Generative AI foundations to single-agent design and coordinated multi-agent systems. With evaluation, guardrails, and human oversight integrated at every stage, you learn to build systems that are both reliable and capable.

With a structured learning experience, the program brings together **learning from experts** and **learning by doing**, critical to developing the intuition and **strategic judgment** needed to design and build agentic AI systems. You engage with **Johns Hopkins University faculty** through recorded lectures and live masterclasses, and with **industry experts** through weekly live mentorship sessions. You apply each concept through **hands-on projects** and **real-world case studies,** using tools like Python, LangChain, and LangGraph.

Upon completion, you earn a Certificate of Completion and 10 Continuing Education Units (CEUs) from Johns Hopkins University, a formal recognition of the technical expertise you develop through the program.

HOW THE PROGRAM HELPS YOU DEVELOP THE SKILLS REQUIRED TO BUILD AGENTIC AI SYSTEMS

Skills required today	Skills this program helps you build
The technical judgment to understand how LLMs work, evaluate their outputs, and diagnose why they fail.	A strong foundation in LLM fundamentals, architecture, and evaluation frameworks to assess model quality and reliability.
The ability to build end-to-end GenAI workflows that go beyond prompting and can retrieve, ground, and evaluate outputs reliably.	RAG-based GenAI pipelines, with the skill to evaluate outputs for accuracy, groundedness, and relevance.
The ability to design Agentic AI systems that plan, use tools, and execute complex workflows.	Single- and multi-agent systems built with LangChain and LangGraph that coordinate toward shared goals and evaluate their own performance.
The ability to deploy AI systems responsibly, with guardrails and oversight that hold up in production.	A portfolio of hands-on projects applying responsible AI principles, including guardrails, human-in-the-loop oversight, and compliance frameworks, to real business problems.



YOUR 14-WEEK AGENTIC AI SKILL JOURNEY THROUGH THIS PROGRAM

Most professionals experiment their way into Agentic AI through a scattered mix of tutorials and demos that never quite builds the depth needed to move past the prototype stage. This program takes you through a clear progression, from Generative AI foundations to designing and evaluating multi-agent systems.

TODAY

You are a technology professional working with Agentic AI prototypes, keen to build end-to-end applications that hold up in production.

HALFWAY THROUGH THE PROGRAM

You can:

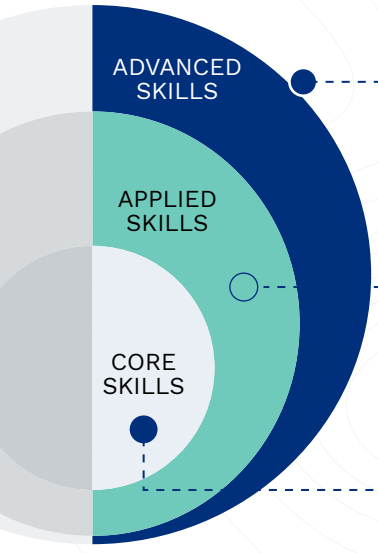
- Build and evaluate GenAI applications grounded in real data using RAG
- Design single-agent systems using the ReAct pattern and MCP
- Evaluate model outputs using LLM-as-a-Judge methods.

AT PROGRAM COMPLETION

You can:

- Design, orchestrate, and evaluate multi-agent systems
- Apply responsible AI guardrails and human-in-the-loop oversight
- Build end-to-end Agentic AI applications with measurable business impact.

Build the Agentic AI skills that compound with time



Advanced Skills
Multi-Agent Orchestration, Responsible AI Guardrails, End-to-End Evaluation

Applied Skills
Single-Agent Systems (ReAct, MCP), RAG Pipeline Design, LLM-as-a-Judge Evaluation

Core Skills
Python and LLM Fundamentals

CORE SKILLS:
The foundation that stays constant, no matter how fast AI evolves.

APPLIED SKILLS:
Becoming the new standard most professionals are still catching up on.

ADVANCED SKILLS:
The hardest skills to self-teach, and the biggest differentiator.

PROGRAM HIGHLIGHTS

Duration: 14 Weeks

Weekly Commitment: 8–10 hours

Format: Delivered online, featuring recorded video lectures, live masterclasses led by Johns Hopkins University faculty, and live mentorship sessions with industry experts.

Hands-On Learning: Build practical expertise by applying Agentic AI to real-world business challenges through projects and case studies, using industry-leading tools with dedicated access to OpenAI labs from Great Learning.

Learning Support: Get personalized assistance from a dedicated Program Support Team, along with academic support through the Great Learning community, project discussion forums, and peer groups.

Compelling e-Portfolio: Build an industry-ready portfolio to showcase your skills and tool proficiency.

Claude-Based AI Workflows Module: Build practical capability in applying Generative and Agentic AI using Claude across real-world workflows.

Recognition: Upon program completion, earn a globally recognized Certificate of Completion and 10 Continuing Education Units (CEUs) from Johns Hopkins University.

EARN A CERTIFICATE OF COMPLETION FROM JOHNS HOPKINS UNIVERSITY



The image is for illustrative purposes only. The actual certificate may be subject to change at the discretion of Johns Hopkins University.

Note: The certificate awarded to learners enrolled in the Code Track will be labeled “Developer Track.”

WHO IS THIS PROGRAM FOR?



Data and AI Practitioners

You work with data, models, and AI/ML solutions. This program helps you create intelligent agentic systems from scratch that can make decisions and adapt to dynamic environments.

For: Data Analysts, Data Scientists, AI/ML Engineers



Technical and Product Managers

You lead teams and shape how business processes run. This program helps you translate current business process workflows into agentic workflows and lead your team in adopting them.

For: Technical Managers, Product Managers



Science, Technology, Engineering, and Mathematics (STEM) Professionals

You have a technical background and are familiar with programming, data, or mathematics. This program helps you develop practical Python skills, work with APIs and frameworks, and build AI agents from the ground up.

For: STEM Professionals, Technology Professionals

Note: Learners with no prior familiarity with technical tools or programming languages can learn Python through a prework journey and assess themselves before the program begins. Based on performance, they may continue directly into the program or take extra weeks to prepare and join the next cohort.

PREFER A NO-CODE PATH?

The **Certificate Program in Agentic AI** is also available as a No-Code Track, designed for business and knowledge professionals. Both tracks share the same conceptual foundation, from Generative AI fundamentals to multi-agent system design, differing only in the tools used to build agentic systems.

The No-Code Track applies the same agentic principles covered in this program through visual, no-code platforms such as n8n, Claude, ChatGPT, and Gemini, making it ideal for professionals who want to apply Agentic AI in their day-to-day work.

TALK TO A PROGRAM ADVISOR



LEARNING OUTCOMES

- Understand Agentic AI, LLM architecture, Retrieval-Augmented Generation (RAG), and reasoning approaches such as ReAct and Chain-of-Thought to differentiate agentic systems from traditional AI and Generative AI.
- Apply agent architectures and classify agent types, including simple-reflex, goal-based, and utility-based, to design single-agent and multi-agent systems with proper coordination patterns.
- Develop functional agents using Python, LangChain, and LangGraph, with integrations across APIs, vector databases, and business systems.
- Assess agent performance across task success, reasoning quality, tool execution, and efficiency, building comprehensive test suites and applying LLM-as-a-Judge evaluation methods.
- Implement guardrails, human-in-the-loop mechanisms, and compliance frameworks to address ethical challenges in designing autonomous AI systems.
- Create end-to-end Agentic AI applications that integrate LLMs, RAG, agent design, multi-agent coordination, and evaluation with clear business impact.



DELIVERED BY WORLD-CLASS FACULTY



Ian McCulloh, Ph.D.
Director, AI Executive & Professional Education, Johns Hopkins University

Dr. Ian McCulloh leads the Artificial Intelligence portfolio for Johns Hopkins Engineering Executive and Professional Education. His pioneering research integrates AI, neuroscience, and human behavior to develop scalable, real-world systems that expand access to essential products, services, and healthcare. Dr. McCulloh has also served as Accenture's Chief Data Scientist, where he built and led Accenture's Federal AI practice. Dr. McCulloh earned his Ph.D. in Computer Science from Carnegie Mellon University and is a prolific author with more than 100 peer-reviewed publications.



Dr. Shelby Wilson
Senior Data Scientist, The Johns Hopkins University Applied Physics Laboratory

Dr. Shelby Wilson is a distinguished Senior Data Scientist at the Johns Hopkins University Applied Physics Laboratory (JHU/APL) and holds a faculty position at Johns Hopkins University Engineering for Professionals. With profound expertise in applied mathematics, mathematical biology, and data science, her work encompasses computational epidemiology, dynamical systems, and machine learning to model complex biological phenomena. Her leadership extends to her role as Associate Director for Education, Training, Outreach, and Participation for the NSF Center for the Analysis and Prediction of Pandemic Expansion (APPEX), where she contributes to national efforts in public health. In recognition of her significant scientific contributions to the field and her commitment to diversity, Dr. Wilson was a recipient of the Presidential Recognition Award from the Association for Women in Mathematics.

Dr. Wilson earned a dual Bachelor of Science degree in Mathematics and Computer Science from Spelman College. She then earned her Ph.D. in Applied Mathematics from the University of Maryland, College Park, where her research focused on mathematical models of immune regulation and cancer vaccines. Her career includes a postdoctoral fellowship at the French National Institute for Research in Digital Science and Technology (INRIA) in Grenoble, France, and faculty positions at both Morehouse College and the University of Maryland before joining JHU/APL. She has over a decade of experience developing and applying mathematical and computational approaches to solve critical research problems.



Dr. William Gray-Roncal
Principal Research Scientist, Johns Hopkins University Applied Physics Laboratory

Dr. William Gray-Roncal is a Principal Research Scientist at the Johns Hopkins University Applied Physics Laboratory, where he leads interdisciplinary research in areas such as neuroscience, artificial intelligence, machine learning, and healthcare. With over 20 years of professional experience as an electrical engineer across diverse domains including space, submarines, and biomedicine, his current projects focus on research to estimate and understand brain networks at the synaptic scale, and to explore innovative approaches for rapid scene analysis and decision-making. His work frequently integrates learning and education research into his science and engineering projects. Through efforts including the College Prep and CIRCUIT programs, he has developed models that identify and support top STEM talent to become leaders in their fields, while simultaneously achieving innovative solutions through student research collaborations. Over the last two decades, he has mentored nearly 500 students.

His research has made significant contributions to the field of connectomics, with expertise spanning diverse areas including computer vision, signal and image processing, machine learning, network analysis, artificial intelligence, and systems engineering. He recently published a paper as part of a Nature special collection highlighting a large-scale research effort to map the structural and functional connectivity of a cubic millimeter of mouse visual cortex at nanoscale resolution. This work has implications for neuroscience understanding, health applications, and next-generation AI algorithms. Dr. Gray-Roncal holds a Ph.D. in Computer Science from Johns Hopkins University, an M.S. in Electrical Engineering from the University of Southern California, and a B.S. in Electrical Engineering from Vanderbilt University. He has academic appointments in Computer Science, Johns Hopkins Engineering Executive and Professional Education within the Whiting School of Engineering, as well as an appointment in the JHU School of Education.



Dr. Iain Cruickshank
Faculty Member, Johns Hopkins University

Dr. Iain Cruickshank holds faculty appointments in Computer Science at Johns Hopkins University and Carnegie Mellon University. He is a practicing data scientist and Artificial Intelligence (AI) engineer for the U.S. Army. During his tenure with the U.S. Army, Cruickshank has been a data scientist at the Army's Artificial Intelligence Integration Center, a senior research scientist at the Army Cyber Institute, and has held various leadership and technical positions within the U.S. Army. His research is focused on novel, machine learning–based techniques for understanding online information environments and platforms, and on the test, evaluation, and acquisition of machine-learning-enabled systems. Techniques from his research have been used by industry and academic research groups to analyze online misinformation spread and state-sponsored disinformation campaigns. He has published more than 40 peer-reviewed academic papers.

Dr. Cruickshank received a Ph.D. in Societal Computing from Carnegie Mellon University, a Master of Science in Operations Research from the University of Edinburgh, and a Bachelor of Science in Mathematics from the United States Military Academy at West Point. His honors include a National Science Foundation Graduate Research Fellowship and a Rotary Ambassadorial Scholarship.



Dr. Pedro Rodriguez
Faculty, Johns Hopkins University AI Program

Dr. Pedro Rodriguez is a faculty member in the Johns Hopkins University Artificial Intelligence Program and leads the Information Science Branch at the Johns Hopkins University Applied Physics Laboratory. He oversees a team of more than 250 AI/ML researchers working on cutting-edge projects for the Department of Defense, the Intelligence Community, and other government agencies.

With more than 20 years of experience, Dr. Rodriguez specializes in AI/ML algorithms for target detection, tracking, classification, and sensor fusion. He holds a Ph.D. in Electrical Engineering from the University of Maryland, Baltimore County; an M.S. in Applied Biomedical Engineering from Johns Hopkins University; and a B.S. in Electrical Engineering from the University of Puerto Rico.

Note: This is an indicative list and is subject to change based on the availability of faculty.

PROGRAM CURRICULUM

PREPARATORY MODULE

This module gets learners set up with all the tools needed for the learning journey, builds a solid foundation in Agentic AI concepts and evolution, and familiarizes learners with development environments and platforms (Python/VS Code).

- The AI and Agentic AI Landscape
- Python Fundamentals
- Applications of AI and Agentic AI (Live Session)
- Tool Setup Session (Live Session)
- Python Essentials for Building Agentic AI Solutions (Live Session)

MODULE 01: GENERATIVE AI FOUNDATIONS

This module covers Large Language Model architecture, prompt engineering techniques (Chain-of-Thought, few-shot, zero-shot), and Retrieval-Augmented Generation (RAG) to ground AI in real data. Learners apply evaluation frameworks to measure and improve system quality, culminating in Project 1, a functional GenAI application solving a real business problem.

Week 01: AI-Assisted Coding with Python

- Understand how AI coding assistants work and explain the “Generate → Review → Refine” workflow for sustainable AI-assisted development.
- Evaluate AI-generated code for correctness, logic errors, and security vulnerabilities; identify when to trust, modify, or rewrite AI code.
- Assess practical applications of AI-assisted coding for rapid prototyping, debugging, and solving Agentic AI problems.

Week 02: LLMs and Prompt Engineering

- Understand how Large Language Models work internally, including tokenization, embeddings, attention mechanisms, and inference, to inform prompt design and model selection.
- Explain core prompting techniques (zero-shot, few-shot, Chain-of-Thought, Tree-of-Thought) and when to apply each for different problem types.
- Evaluate trade-offs between models in terms of capability, speed, and cost; select appropriate models for specific tasks.
- Demonstrate the ability to optimize prompts iteratively and implement prompt management at scale using Python APIs.

Week 03: Retrieval-Augmented Generation (RAG) and Advanced RAG

- Understand the RAG pipeline architecture, including document ingestion, chunking, embedding, vector storage, retrieval, and generation, to ground LLMs in real data.
- Explain trade-offs between chunking strategies (fixed-size, semantic, overlapping) and vector databases (ChromaDB, Pinecone, FAISS) on retrieval quality and speed.
- Assess advanced RAG techniques, including hybrid search, query expansion, reranking, and knowledge graphs, and determine which to apply for specific use cases.
- Demonstrate the ability to build and evaluate end-to-end RAG pipelines using LangChain, vector databases, and RAGAS evaluation frameworks.

Week 04: Prompt Optimization and Evaluation

- Understand evaluation metrics and frameworks (RAGAS, DeepEval, DSPy, LLM-as-a-Judge) to measure GenAI system quality across dimensions such as accuracy, hallucination rate, and consistency.
- Assess hallucination detection and mitigation strategies, including identifying false claims and improving LLM reliability through prompt refinement and RAG grounding.
- Evaluate trade-offs between models through A/B testing; design iterative evaluation loops (baseline, test, score, refine) for continuous improvement.
- Demonstrate the ability to build comprehensive test sets, implement evaluation judges, and track improvements using quantitative metrics and monitoring.

WEEK 05 — PROJECT 1

WEEK 06 — LEARNING BREAK



MODULE 02: INTRODUCTION TO AGENTIC AI DESIGN

This module transitions from responsive to autonomous AI, exploring agent architectures and building single-agent systems. Learners design agents that perceive, reason, plan, and act using industry frameworks and platforms. The module embeds safety, ethics, and responsible AI practices throughout, culminating in Project 2, a coordinated multi-agent system solving complex challenges.

Week 07: Core Concepts of Agentic AI Systems

- Understand what makes an agent "agentic," including autonomy, goal-directed behavior, reasoning, planning, tool use, and learning capabilities.
- Explain different agent architectures and classify agent types (simple-reflex, goal-based, utility-based) to determine which fits specific problem types.
- Assess memory systems for agents, including short-term (context), long-term (retrieval), episodic, and semantic, and design appropriate memory strategies.

Week 08: Single-Agent Systems

- Understand the ReAct (Reasoning + Acting) pattern and Model Context Protocol (MCP) for agent decision-making and tool integration.
- Design and implement a complete single-agent system for an end-to-end task using LangChain, LangGraph, and agent frameworks.
- Build agents with custom tools, memory systems, and error recovery; manage state transitions and agent behavior.
- Evaluate agent performance using task-completion metrics, reasoning trajectory analysis, and LLM-as-a-Judge methods; identify areas for improvement.

Week 09: Ethics, Safety, Alignment and Responsible AI

- Understand the AI alignment problem and explain agentic-specific risks, including specification gaming, reward hacking, negative side effects, and autonomous decision-making failures.
- Apply responsible AI principles (fairness, transparency, accountability, privacy) to agent design; identify potential bias, discrimination, and ethical concerns.
- Assess safety techniques, including rule-based guardrails, neuro-symbolic approaches, human-in-the-loop mechanisms, monitoring, and anomaly detection.
- Demonstrate the ability to implement guardrails in agent code, design evaluation for safety compliance, and document governance and audit trails.

WEEK 10 — PROJECT 2

WEEK 11 — LEARNING BREAK



MODULE 03: BUILDING AGENTIC AI SYSTEMS

This module covers sophisticated multi-agent orchestration and rigorous evaluation frameworks. Learners design comprehensive test suites, implement governance and compliance, and validate system behavior end-to-end, culminating in Project 3, an end-to-end agentic solution addressing complex organizational challenges.

Week 12: Multi-Agent Systems

- Understand Multi-Agent Systems (MAS) architecture and explain coordination challenges when multiple agents interact and collaborate.
- Compare multi-agent architectures, including hierarchical (manager-worker), flat/peer-based, market-based, and conversational, and determine which fits specific business problems.
- Design Agent-to-Agent (A2A) communication patterns; implement data hand-offs, message passing, and conflict resolution between specialized agents.
- Demonstrate the ability to build collaborative multi-agent systems using LangGraph; coordinate agents toward shared goals and manage system state.

Week 13: Evaluation of Agentic AI Systems

- Understand evaluation dimensions for agents, including task success, reasoning quality, tool execution, efficiency, and safety compliance.
- Apply evaluation methodologies (LLM-as-a-Judge, rule-based testing, human-in-the-loop review) to assess multi-agent system behavior and performance.
- Design comprehensive test suites covering happy path, edge cases, adversarial scenarios, and regression testing for agent robustness.
- Demonstrate the ability to build neuro-symbolic evaluation judges, identify system bottlenecks, and propose optimizations based on evaluation results.

WEEK 14 — PROJECT 3

SELF-PACED MODULES

FOUNDATIONS OF AI/ML

Learn how the AI lifecycle works, from framing a business problem to deploying and monitoring a model, and compare core algorithms such as Logistic Regression, SVM, Decision Trees, Random Forests, and Neural Networks.

NLP FOUNDATIONS AND TRANSFORMERS

Learn how machines process language, from tokenization and TF-IDF to Word2Vec and modern embeddings. See how the Transformer architecture and self-attention replaced RNNs and LSTMs to power today's LLMs.

AI-ASSISTED CODING FOR ML WORKFLOWS

Build prediction models with Codex in VS Code, using a Generate, Review, Refine workflow. Work through exploratory data analysis, leakage detection, preprocessing pipelines, and baseline model evaluation, while retaining full ownership of every implementation decision.

FINE-TUNING SMALL LANGUAGE MODELS

Understand when a small language model (SLM) is a better choice than a large one, and fine-tune an SLM to reduce cost and latency. This module pairs well with multi-agent systems, where SLMs can power fast, low-cost sub-agents.

REINFORCEMENT LEARNING

Learn the core reinforcement learning concepts of agent-environment loops, rewards, and policies, and compare value-based and policy-based algorithms. Judge when an agent should learn its behavior and when it should simply follow instructions.

CLAUDE-BASED AI WORKFLOWS

This module is designed to build practical capability in applying Generative AI and Agentic AI using the Claude ecosystem in real-world contexts. Learners build the ability to design, execute, and evaluate AI-driven workflows for real-world applications, supported by ~5 hours of structured learning.

Design and Execute AI Workflows

- Model selection and Prompt Engineering using **Claude Chat**
- Agentic workflow design and orchestration using **Claude CoWork**
- Plan → Approve → Execute → Iterate framework
- Designing workflows with reasoning, tools, and multi-step execution
- Applying concepts through real-world case studies

Build and Deploy AI Systems at Scale

- API integration and model usage using **Claude Code**
- Tool integration using the Model Context Protocol
- Designing agentic systems with memory, tools, and orchestration
- Performance optimization, cost considerations, and system reliability
- Responsible AI principles, including alignment approaches such as Constitutional AI

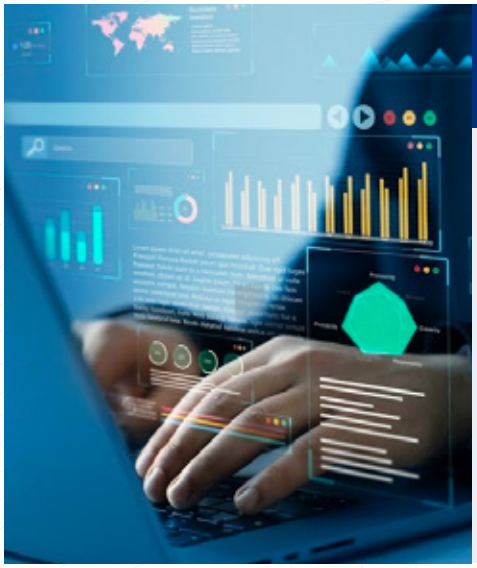
***Disclaimer:** Claude API access is not included as part of the program. Learners may explore advanced usage independently.

ANTHROPIC MASTERCLASS

This masterclass covers the Anthropic AI landscape, exploring Claude models, Constitutional AI, and key safety and alignment principles. Learners apply effective prompt engineering techniques, use the Claude API for tasks and integrations, generate structured outputs, build simple applications, compare Claude with other AI models, and evaluate ethical considerations for deploying AI systems.

Note: The curriculum is subject to periodic review and updates at the discretion of the faculty to ensure alignment with industry requirements.

SAMPLE PROJECTS



DUALLENS ANALYTICS — AI-POWERED INVESTMENT RESEARCH ASSISTANT

INDUSTRY: FINANCE, INVESTMENT MANAGEMENT

Objective: Build an AI-powered investment research system that combines market data and company documents to generate ranked investment recommendations with confidence scores. Learners implement RAG for document grounding, LLM-as-a-Judge for quality evaluation, DSPy GEPA for prompt optimization, and quantitative-qualitative analysis to deliver trustworthy, actionable intelligence.

Skills you will learn: LangChain, RAG, Prompt Optimization, LLM-as-a-Judge, DSPy, Structured Reasoning



SENTINEL FINANCE — FINANCIAL INTELLIGENCE AGENT

INDUSTRY: FINANCE, INVESTMENT ANALYSIS

Objective: Build a financial intelligence agent that integrates real-time market data, news sentiment, private reports, and MCP-connected tools to deliver explainable investment analysis. Learners combine RAG for document grounding, multi-agent coordination, responsible AI principles, and LLM-as-a-Judge evaluation to create a trustworthy financial decision-support system.

Skills you will learn: LangChain, LangGraph, RAG, MCP, Responsible AI, LLM-as-a-Judge, Real-Time Data Integration



CLINICAL EVIDENCE SYNTHESIS AGENT FOR RESEARCH TEAMS

INDUSTRY: HEALTHCARE, PHARMACEUTICAL RESEARCH

Objective: Build a multi-agent research assistant that pulls literature from PubMed and clinical trial registries, combines it with a private library of study protocols and internal reports, and uses MCP-connected tools to produce structured, source-cited evidence summaries. Learners use RAG to ground answers in documents, coordinate agents for search, screening, extraction, and synthesis, and apply responsible AI principles such as clear disclaimers, bias and conflict-of-interest flags, and a rule that the system never gives patient-specific medical advice. Evaluation metrics are used to verify that each claim is traceable to a source and that the summary does not overstate study findings.

Skills you will learn: LangChain, LangGraph, RAG, MCP, Responsible AI, LLM-as-a-Judge, Real-Time Data Integration, DeepEval's Agent Evaluation Metrics



CONTRACT RISK REVIEW AGENT

INDUSTRY: LEGAL, COMPLIANCE

Objective: Build a multi-agent system that reviews contracts against a legal playbook and regulatory updates. Agents use RAG to ground each finding in the contract text, and MCP tools for document and ticketing systems. Learners evaluate the system using DeepEval agent metrics, including Task Completion and Tool Correctness for the workflow, Faithfulness and Contextual Relevancy for retrieval grounding, and Hallucination and PII Leakage for safety. A human lawyer makes the final call.

Skills you will learn: LangChain, LangGraph, RAG, MCP, Responsible AI, LLM-as-a-Judge, Real-Time Data Integration

Note: This is an indicative list of projects and is subject to change.



SAMPLE CASE STUDIES



GLOBAL SUPPLY CHAIN DISRUPTION IMPACT ASSESSMENT

Industry: Supply Chain and Logistics

Objective: Transform structured supply chain disruption data into executive-ready business reports by building a Supply Chain Impact Assessment system using LangChain and LLM prompting techniques. Learners compare Zero-Shot, Few-Shot, Chain-of-Thought, and Tree-of-Thought strategies while implementing prompt parameter tuning to generate stakeholder-specific insights and evaluate the impact of different prompting approaches on report quality.

Skills you will learn: LangChain, Prompt Engineering, Zero-Shot, Few-Shot, CoT, ToT, Prompt Parameter Tuning



DOCUMENT Q&A USING RETRIEVAL-AUGMENTED GENERATION

Industry: Finance

Objective: Enable question-answering capabilities over business documents by building an end-to-end RAG pipeline that ingests PDFs, chunks content, generates embeddings, and retrieves relevant context. Learners implement LLM-as-a-Judge evaluation to measure groundedness, context relevance, and answer relevance, ensuring the reliability and trustworthiness of RAG applications.

Skills you will learn: RAG, LangChain, ChromaDB, LLM-as-a-Judge, Embeddings



AI-POWERED CUSTOMER REVIEW SUMMARIZATION FOR NOVATECH

Industry: Product Intelligence, E-commerce

Objective: Convert thousands of customer reviews into actionable, structured summaries for Product Managers by building an evaluated AI pipeline that progresses from naive baseline to zero-shot structured prompting, few-shot prompting, and GEPA prompt optimization. Learners evaluate each version on faithfulness, sentiment balance, and actionability, using business confidence gates to determine auto-approval, human review, or rejection thresholds.

Skills you will learn: DeepEval, Prompt Optimization, GEPA, ROUGE-L, Evaluation Frameworks, Pandas



AGENTIC RAG FOR COMPETITIVE INTELLIGENCE

Industry: Consulting, Market Research

Objective: Build an intelligent competitive intelligence agent using LangGraph that validates company information, identifies sectors and competitors, and retrieves insights from internal knowledge bases. The agent dynamically routes to web search when internal knowledge is insufficient and synthesizes structured competitive analysis reports through conditional graph execution and self-correction.

Skills you will learn: LangGraph, Agentic RAG, LangChain, ChromaDB, Agent Reasoning, Dynamic Routing

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LEXAGENT — AI-POWERED RENTAL LAW REASONING SYSTEM

Industry: LegalTech

Objective: Build LexAgent, a legal reasoning assistant that interprets rental regulations, retrieves relevant legal documents and case law, reviews lease agreements for compliance, and explains tenant and landlord rights. Learners combine LangGraph, Model Context Protocol (MCP), and RAG to create a jurisdiction-aware legal assistant and evaluate agent performance using task-completion metrics and LLM-as-a-Judge.

Skills you will learn: LangGraph, ReAct, MCP, RAG, Legal Document Retrieval, LLM-as-a-Judge

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RESPONSIBLE AI FOR AUTONOMOUS RETURNS AND REFUNDS AGENT

Industry: E-commerce, Financial Services



Objective: Build a secure, trustworthy e-commerce returns and refunds agent using LangGraph and MCP while implementing guardrails against adversarial attacks (prompt injection, PII exfiltration, toxicity, over-refund attempts). Learners implement a neuro-symbolic architecture with input/output safety checks, conditional routing, and policy enforcement to balance automation with responsible AI governance.

Skills you will learn: LangGraph, ReAct, MCP, Safety Guardrails, Responsible AI, Presidio, Detoxify, Neuro-Symbolic Reasoning



AI RESEARCH MULTI-AGENT SYSTEM

Industry: Research, Academia

Objective: Automate the end-to-end research workflow by building a collaborative multi-agent system where specialized agents retrieve academic literature, analyze evidence, synthesize findings, and generate structured research reports. Learners implement Agent-to-Agent communication using LangGraph and MCP, demonstrating how coordinated agents solve complex research tasks more effectively than single agents.

Skills you will learn: LangGraph, Multi-Agent Systems, A2A Communication, MCP, LangChain, Collaborative Reasoning



HEALTHCARE INTELLIGENCE ASSISTANT WITH HUMAN-IN-THE-LOOP

Industry: Healthcare, Medical Intelligence

Objective: Build a healthcare assistant that converts natural-language questions into SQL queries over healthcare databases while maintaining human oversight for critical decisions. Learners implement LangGraph interrupts and state persistence to route READ queries for auto-execution, WRITE operations to human approval gates, and block unsafe operations, maintaining audit trails for compliance and review.

Skills you will learn: LangGraph, Human-in-the-Loop, SQL Query Generation, State Management, Audit Trails, Safety Gates



TRIAGE BENCH — EVALUATING PLANNING AND SELF-CORRECTION IN SUPPORT AGENTS

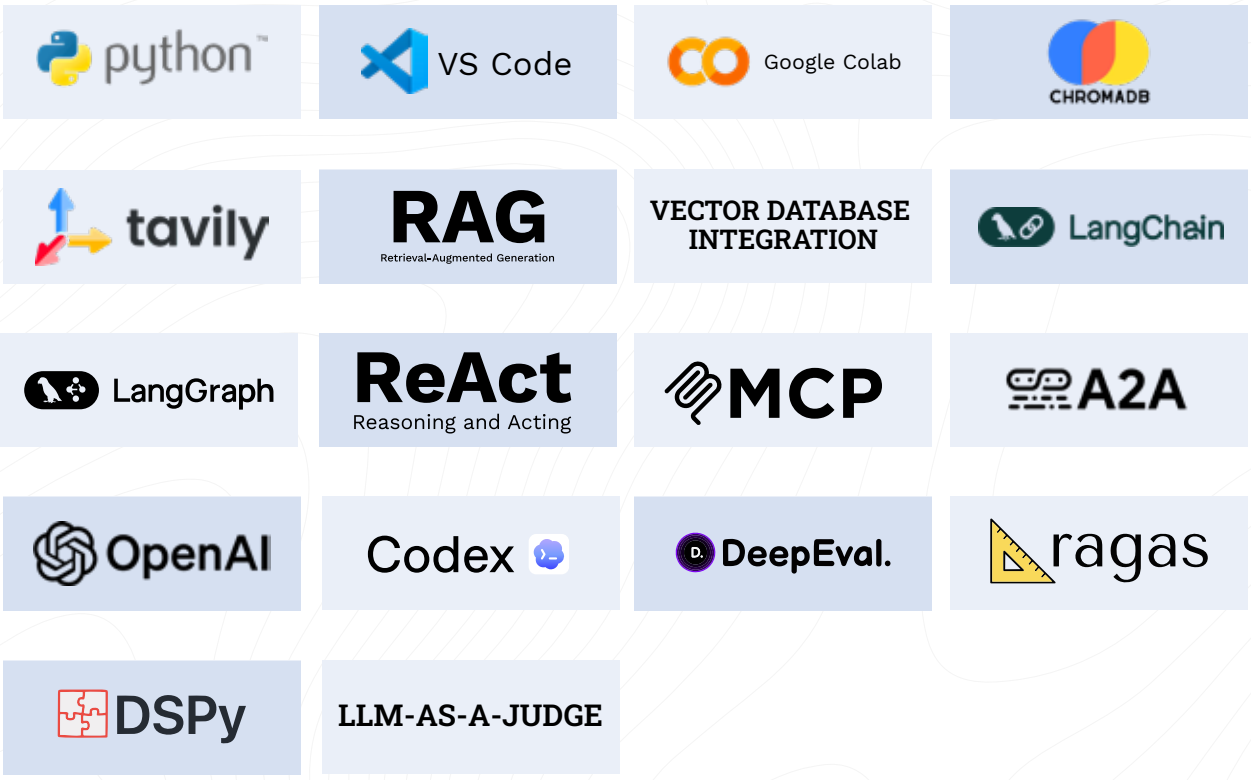
Industry: Customer Support, Operations

Objective: Evaluate whether planning and self-correction improve customer support ticket triage quality by comparing two LangGraph agents on the same frozen golden dataset. One agent uses reactive routing; the other uses planning, structured arguments, and self-correction. Learners evaluate performance across reasoning, action, and execution metrics using DeepEval and LLM-as-a-Judge.

Skills you will learn: LangGraph, DeepEval, Agent Evaluation, Planning vs. Reactive Approaches, LLM-as-a-Judge

Note: This is an indicative list of case studies and is subject to change.

TOOLS AND TECHNOLOGIES



Note: This is an indicative list, and relevant tools are reviewed for inclusion periodically. Access to certain tools may require separate subscriptions or fees.



INDUSTRY MENTORS



BHASKARJIT SARMAH
Head of AI Research
at Domyn



BRIDGET HUANG
GenAI/ML Engineer
at Amazon Web
Services (AWS)



KALLE BYLIN
Data Engineer at
Workday



TANYA GLOZMAN
Applied Science -
AI/ML at Apple



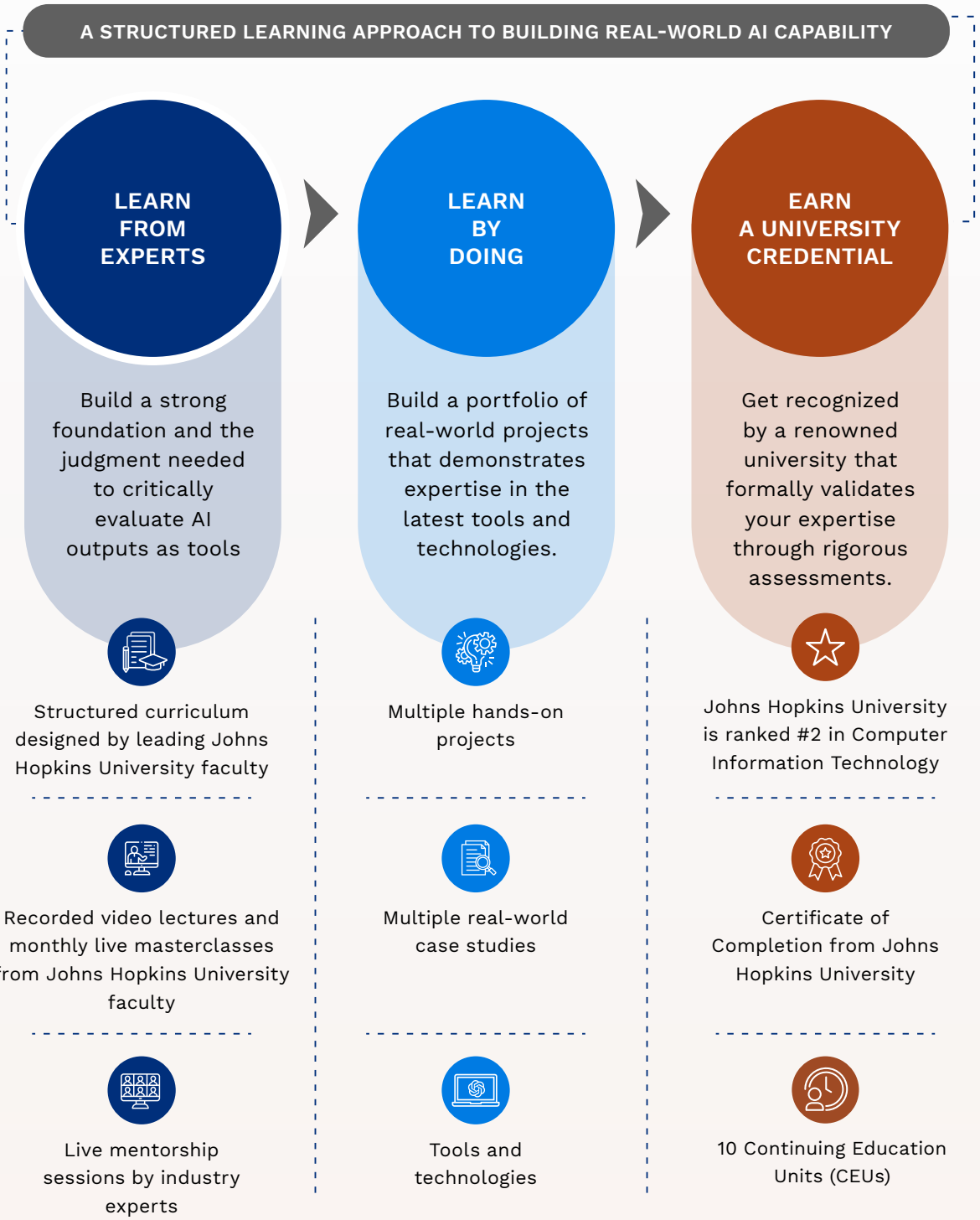
VINICIO DE SOLA
Staff Data Scientist
at Newmark

Note: This is an indicative list and is subject to change based on the availability of mentors.

STRUCTURED LEARNING IN THE AGE OF AI

Today, the technical knowledge barriers to learning and using AI have diminished greatly. However, barriers of a different kind that impact its successful implementation have emerged; those involving **human intuition and strategic judgment.** To consistently get the best from any AI tool and drive business impact, one must understand the fundamentals of the underlying technology.

Our structured approach combines **learning from experts** who have created the curriculum and **learning by doing** through hands-on work using the latest tools and technologies. Together, these ensure a well-rounded learning experience. **The university credential** provides formal validation of your expertise, a valuable differentiator in a world where everyone claims to be proficient in AI.



KEEP YOUR LEARNING R.E.A.L. WITH OUR STRUCTURED LEARNING PROGRAMS

As technology evolves, self-paced learning offers a convenient way to stay updated on individual tools. But knowing how to use a tool is only part of what it takes to deploy a working solution. That requires an understanding of the underlying concepts, how the components work together, and why they fail in practice. For professionals seeking to create enterprise impact, this is where the depth and guided application of structured learning matter.



R | **Rigorous Foundations:** Develop a deep understanding of core concepts and how they connect, from foundations to application.



E | **Expert Instruction:** Learn from faculty and industry experts to develop technical intuition grounded in real-world experience.



A | **Applied Learning:** Work on real-world, hands-on projects and receive expert feedback.



L | **Learning for Impact:** Build the judgment to translate learning into measurable outcomes at work.

WHAT IS BUILT INTO YOUR LEARNING EXPERIENCE

Feature	Self-Paced Learning	GL Structured Programs
Learn how to use	✓	✓
Stay updated on new tools and technologies	✓	✓
Learn with convenience and flexibility	✓	✓
University-backed certificate and industry-focused curriculum designed by leading academicians	×	✓
In-depth programs with comprehensive, end-to-end coverage	×	✓
Real-world, hands-on projects with expert feedback	×	✓
Weekly live interactions with industry mentors to build practical judgment	×	✓
Program Support Team for guidance, tracking, and completion	×	✓



ENROLLMENT PROCESS

The enrollment process is conducted on a rolling basis and will be closed once the requisite number of candidates is enrolled.

APPLY
Fill out an
online application form

REVIEW
Eligible applications will be reviewed
on a rolling basis

JOIN PROGRAM
An offer letter will be sent to
selected candidates

PROGRAM FEE

1,75,000 + GST

Please contact the Program Advisor from Great Learning for more information on offers, payment plans, and eligibility for financial assistance.

PROGRAM PARTNER

Great Learning
POWER AHEAD

Great Learning is a leading global ed-tech company for professional and higher education. It offers comprehensive, industry-relevant programs across cutting-edge Technology, Data, and Business domains. These programs are developed in collaboration with the world's foremost academic institutions, such as Johns Hopkins University, MIT Professional Education, The University of Texas at Austin, Northwestern School of Professional Studies, Deakin University, Chicago Booth Executive Education, Indian Institute of Technology Bombay, S. P. Jain Institute of Management and Research, Great Lakes Institute of Management, and more.

Programs offered by Great Learning are constantly reimagined and revamped to address the dynamic needs of the industry. Delivered in blended, classroom, and purely online modes, these programs are supported by expert mentors and highly qualified faculty. Great Learning is on a mission to enable transformative learning and career success in the digital economy and has impacted 16 million+ learners from over

16 Mn+
Learners

170+
Countries

8600+
Industry Experts



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